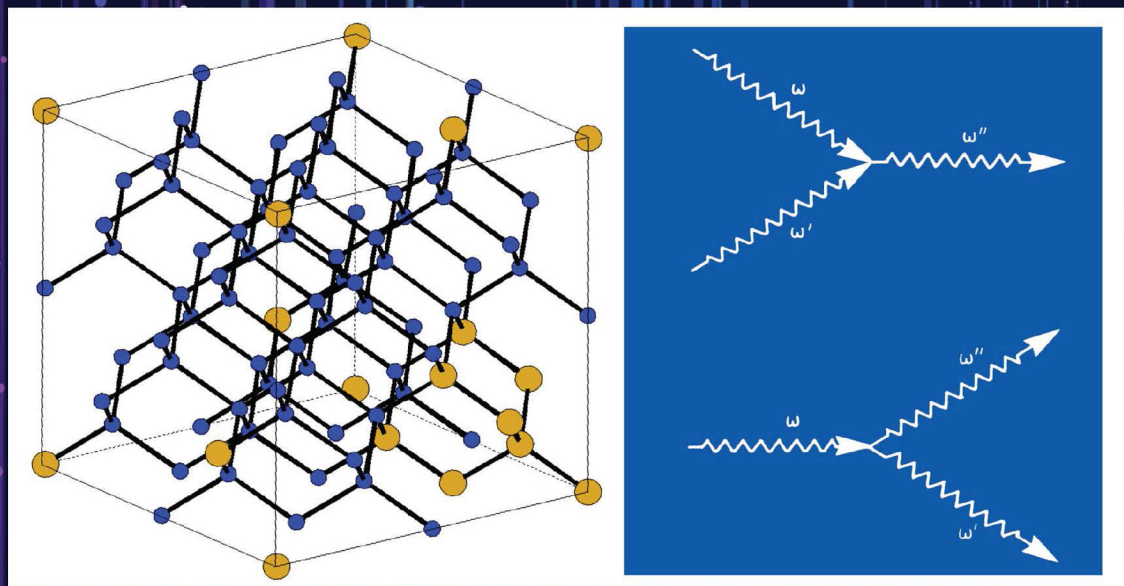


The Physics of Phonons

SECOND EDITION



Gyaneshwar P. Srivastava



CRC Press
Taylor & Francis Group

The Physics of Phonons

This fully updated second edition of **The Physics of Phonons** remains the most comprehensive theoretical discussion devoted to the study of phonons, a major area of condensed matter physics.

It contains exciting new sections on phonon-related properties of solid surfaces, atomically thin materials (such as graphene and monolayer transition metal chalcogenides), in addition to nanostructures and nanocomposites, thermoelectric nanomaterials, and topological nanomaterials, with an entirely new chapter dedicated to topological nanophononics and chiralphononics. Although primarily theoretical in approach, the author refers to experimental results wherever possible, ensuring an ideal book for both experimental and theoretical researchers.

The author begins with an introduction to crystal symmetry and continues with a discussion of lattice dynamics in the harmonic approximation, including the traditional phenomenological approach and the more recent *ab initio* approach, detailed for the first time in this book. A discussion of anharmonicity is followed by the theory of lattice thermal conductivity, presented at a level far beyond that available in any other book. The chapter on phonon interactions is likewise more comprehensive than any similar discussion elsewhere. The sections on phonons in superlattices, impure and mixed crystals, quasicrystals, phonon spectroscopy, Kapitza resistance, and quantum evaporation also contain material appearing in book form for the first time. The book is complemented by numerous diagrams that aid understanding and is comprehensively referenced for further study. With its unprecedented wide coverage of the field, **The Physics of Phonons** is an indispensable guide for advanced undergraduates, postgraduates, and researchers working in condensed matter physics and materials science.

Features

- Fully updated throughout, with exciting new coverage on graphene, nanostructures and nanocomposites, thermoelectric nanomaterials, and topological nanomaterials.
- Authored by an authority on phonons.
- Interdisciplinary, with broad applications through condensed matter physics, nanoscience, and materials science.



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Vidyāṃ cā'vidyāṃ ca yastadvedobhayam saha.
Avidyayā mṛtyum tīrtvā vidyayā'mṛtamaśnute.

(Yajurveda, XL.14).

(He who knows the nature of knowledge and nescience simultaneously passes over death by means of nescience and attains salvation by the help of true knowledge.)

Dedicated to the memories of my parents, and aunty and uncle Harish



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Preface (Second Edition)

Significant advances have been made in developing concepts, techniques and applications of the physics of phonons since the publication of the first edition of this book in early 1990. In this edition I have attempted to include some aspects of new developments in theoretical and computational studies of phonon physics. These include topics, among others, such as (1) lattice dynamics within the harmonic approximation using the density functional perturbation theory (DFPT), (2) anharmonic phonon interactions (three-phonon and four-phonon processes) using fully *ab initio* treatments and a semi-*ab initio* treatment, (3) *ab initio* treatment of phonon-electron interaction, (4) nanophononics, (5) topological phonons, (6) topological nanophononics, (7) topological phonon chirality and (8) anisotropic effective medium theory for lattice thermal conductivity of nanocomposites.

Chapter-wise, the main additional material in edition 2 is as follows. [Chapter 2](#) deals with lattice dynamics of two-dimensional mesh and provides a discussion on empirical many-body interatomic potential. [Chapter 3](#) has been substantially rewritten, detailing different approaches for *ab initio* treatment of lattice dynamics at harmonic level. [Chapter 4](#) presents semi-*ab initio* and *ab initio* formalisms for cubic and quartic anharmonicity. The semi-*ab initio* approach has allowed for the presentation of explicit expressions for cubic and quartic anharmonic potential terms. [Chapter 6](#) describes methods of evaluation of phonon-phonon and phonon-electron interaction rates using *ab initio* and semi-*ab initio* treatments. A simplified treatment of phonon relaxation arising from phonon-spin interaction has been added. [Chapter 7](#) presents a spectrum of three-phonon relaxation rates and provides a description of thermal conductivity calculation using different levels of relaxation time theory (*e.g.* single-mode, Callaway, iterative) and different schemes for Brillouin zone integration (*e.g.* Debye's isotropic continuum scheme, special wave-vector scheme using phonon eigensolutions from *ab initio* methods). [Chapter 8](#) discusses *ab initio* results for phonons on semiconductor surfaces and for mono-layer thin 2D systems. [Chapter 9](#) presents theory and results for thermal transport in periodically structured nanocomposite. This is done at two levels. For ultrathin nanocomposites (periodicity typically less than 10 nm) the theories described in [chapters 3–7](#) have been used. A fully anisotropic effective medium theory (incorporating anisotropy of each of insert, host and insert-host parts) has been developed and applied for calculation of thermal conductivity of nanocomposites with periodicity larger than the lower end of the spectrum for phonon mean free path (typically larger than 10 nm). Modern developments of the topics of topological nanophononics and topological phonon chirality have been presented in [chapter 10](#). Three new appendices have been added, and a few minor corrections and additions have also been made throughout the book. On the whole, more than 35% new material has been added to edition 1 to prepare edition 2.

Several national and international colleagues have indicated that they would like me to produce a second edition of this book. Also, my family members (wife Kusum, daughter Jyoti, son Deepak, daughter-in-law Marina and granddaughter Elizabeth) have been encouraging me to undertake this project. I must say that the work on this edition would not have progressed smoothly during the present COVID-19 pandemic period had it not been for the huge support from my wife Kusum. For this, I am very grateful to her.

I thank Mrs Rebecca Hodges-Davies, Commissioning Editor for Physics, CRC Press, Taylor & Francis Group, for inviting and supporting me to prepare this edition of the book. I also thank her team members and the desk editor for their help with the technical side of the publication.

It is pleasing that since the publication of the first edition I have been receiving encouraging emails and letters from graduate students and academics from different countries. Also, several good and encouraging reviews of the first edition were published in 1990s. Similarly, there have been helpful reviews of the proposal for the inclusion of additional topics in the second edition. I thank

all the reviewers of the first edition and the proposed second edition of the book and hope that my effort in preparing this edition will meet readers' expectations. In particular, I hope that this edition will prove useful to undergraduates, postgraduates, postdoctoral fellows, experienced researchers as well as academics interested in condensed matter physics, material science and computational condensed matter physics.

Several postgraduates, postdoctoral fellows and collaborators, particularly Professor Hüseyin Tütüncü, Dr Steve Hepplestone and Dr Iorwerth Thomas, have contributed to my knowledge of the subject matter added to this edition. I am truly thankful to all those. I especially thank Dr Iorwerth Thomas for his collaboration in the past decade on thermal transport and for providing useful feedback on a few chapters added to this edition. I would also like to thank Dr S. Raj Vatsya of NRC Canada for useful discussions in 2006 on properties of linear operators, in particular the anharmonic phonon collision operator used in this book. Any misconceptions and mistakes in this edition are failings on my part, and I would welcome suggestions for improvement from readers. It is important to add that many important works and citations are inadvertently missing in this limited-sized book, for which I express my sincere apology. I also apologise for inadvertently using a few symbols to represent more than one variable or physical quantity, but hope that this does not cause much inconvenience to readers.

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February 2022

Acknowledgments (Second Edition)

I am grateful to the following for granting permission to reproduce figures added to the second edition of this book.

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Preface (First Edition)

This book is an outcome of my research interest and teaching at the undergraduate and postgraduate levels. I hope that it will be useful to final-year undergraduates, postgraduates and researchers who seek to study the physics of phonons beyond the level presented in most textbooks currently available on the subject.

I have covered three basic topics of phonon physics in non-metallic solids and liquid helium: (i) vibrational properties, (ii) phonon interactions and (iii) thermal properties. A brief description of the various chapters is as follows.

[Chapter 1](#) presents a brief discussion of the elements of crystal symmetry. [Chapter 2](#) describes the *semiclassical* concepts of lattice dynamics of a linear chain and of three-dimensional crystals. Calculations of the density of normal modes, specific heat and velocity of elastic waves in crystals are also discussed. In [chapter 3](#) the *ab initio* treatment of lattice dynamics in the harmonic approximation is presented at three different levels of sophistication. [Chapter 4](#) introduces the second-quantised notation and derives expressions for the phonon Hamiltonian up to the third-order anharmonic term for both a crystalline solid and an elastic continuum. [Chapter 5](#) describes the theory of lattice thermal conductivity based on three different methods: (i) relaxation-time approaches, (ii) complementary variational approaches and (iii) linear-response approaches. The kinetic theory expression for phonon conductivity, based on the single-mode relaxation-time result, is shown to arise from the diagonal part of the anharmonic phonon collision operator. It is also shown that the classical work of Callaway on thermal conductivity can be understood in terms of an effective phonon relaxation time which includes a contribution from the off-diagonal three-phonon N -processes operator. Furthermore, Callaway's scheme for an effective phonon relaxation time is extended to include the effect of the off-diagonal part of the three-phonon U -processes operator. The theory of complementary variational principles derives a sequence of lower bounds and a sequence of upper bounds, both converging monotonically towards the exact thermal conductivity. This follows among others the work of Arthurs, Benin and the present author. The Green–Kubo linear response expression for thermal conductivity is evaluated by using two different approaches: the double-time Green's function method and the Zwanzig–Mori projection operator method. Although these two approaches are only used here to evaluate the heat current–heat current correlation function, it is hoped that the details of the schemes given in this section will prove useful when applying the linear-response method to other relevant problems. [Chapter 6](#) discusses phonon scattering mechanisms in insulators and doped semiconductors. Results for phonon relaxation times and matrix elements of the phonon scattering operator are derived by using the linear-response and time-dependent first- and second-order perturbation theories, as appropriate. Particular attention has been paid to three-phonon processes involving both acoustic and optical phonons. A brief discussion of the theory of phonon–photon interaction, leading to infrared absorption and Raman scattering processes, is also presented.

[Chapter 7](#) presents numerical results, and their comparison with experiment, for phonon relaxation rates and thermal conductivity of insulators and semiconductors. The theory of phonons on surfaces and in superlattices, using both the continuum model and the crystal model, is presented in [chapter 8](#). This chapter also discusses the interaction of phonons with a two-dimensional gas. [Chapter 9](#) describes the theory of Elliot and Dawber for localised vibrational modes in semiconductors, and the modified-random-element-isodisplacement model for long-wavelength optical phonons in mixed crystals. Also, this chapter discusses infrared results for localised modes in single crystals and Raman scattering results for mode behaviour in mixed crystals. [Chapter 10](#) discusses the physics of phonons in quasicrystals and in amorphous solids. The concept of quasicrystals, their structure and phonon dispersion in the Fibonacci monatomic linear chain is discussed. Recent views on the

nature of vibrational excitations in amorphous solids, namely, phonons, tunnelling states and fractons, and their contribution to the specific heat and thermal conductivity are discussed. [Chapter 11](#) attempts to describe some of the modern phonon spectroscopic techniques for generating and detecting phonons in crystalline semiconductors and in liquid helium. Finally, [chapter 12](#) provides a brief discussion on phonons and rotons in superfluid helium, interaction between these excitations, Kapitza resistance and quantum evaporation.

Each chapter includes references to many original works and reviews which can be consulted to give the reader in-depth knowledge. The list of references is not meant to be exhaustive and I apologise if I have left out any important works.

Some knowledge of undergraduate solid state physics, including electronic band structure, and group theory will be useful in following the subject matter presented in the book.

This book uses a number of unit systems, e.g. CGS and AU; I have not converted these to SI units. Hopefully this will not cause too much confusion to readers.

A few acknowledgments are in order. I am grateful to Professor G S Verma who introduced me to the field of phonon interactions in solids. My sincere thanks are due to all those with whom I have had discussions and collaborations on the subject matter over the years. I am particularly grateful to Professor J E Parrott, Professor L J Challis, Professor D L Weaire, Professor J C Inkson, Professor A F G Wyatt, Dr R Jones, Dr R T Phillips, Dr C Lambert, Dr D Kechrakos, Dr Z Ikonić, Mr P Briddon and Mr P Mulheran who kindly read various chapters of this book and made very valuable suggestions for improvement. However, all remaining misconceptions and mistakes in the book are my responsibility, and I would appreciate hearing from readers about any corrections and suggestions for improvement that could be included in any future edition.

My special thanks go to Mr C S G Cousins whose initial expert advice helped me successfully prepare this manuscript using $\text{\TeX}$.

I would also like to express my sincere thanks to Mr J A Revill, Commissioning Editor for IOP Publishing, and his staff for their interest and support in the publication of this book.

Finally, it could not have been possible to complete this book had I not received the continuous and full support of my wife Kusum and our children Jyoti and Deepak. To them I can only give my love and best wishes in life.

G P Srivastava

Exeter

November 1989

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1 Elements of Crystal Symmetry

In this chapter an introduction to crystal symmetry is presented. Some of the results will be required for the rest of the book. The discussion is intended to be brief, and the reader is recommended to consult further literature for more details. References used in preparing the presentation in this book are, among others, Heine (1960), Bhagavantam (1966), Morgan (1969), Mariot (1972), Boardman *et al* (1973), Callaway (1974), Joshi (1982), Burns (1985) and Kittel (1986).

1.1 DIRECT LATTICE

A *lattice* is defined as an infinite regular array of *points* in which all points have identical surroundings. As a point is a mere realisation, a lattice is just a mathematical construction. From a chosen point, the position vector of another point of a three-dimensional lattice is given by a *translation vector*

$$\mathbf{T} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3, \quad (1.1)$$

where n_1, n_2 and n_3 are any integers, and $\mathbf{a}_1, \mathbf{a}_2$ and $\mathbf{a}_3$ are three independent *primitive translation vectors*. The parallelepiped formed by $\mathbf{a}_1, \mathbf{a}_2$ and $\mathbf{a}_3$, which contains one lattice point, is called a *primitive unit cell*. As all points have identical surroundings, the whole space can be generated by suitable translations of the primitive unit cell. Therefore, the essential features of a lattice can be extracted from its primitive unit cell.

A cell or parallelepiped which contains more than one lattice point is called a *non-primitive unit cell*.

A primitive unit cell in the shape of a parallelepiped defined by $\mathbf{a}_1, \mathbf{a}_2$ and $\mathbf{a}_3$ has a volume equal to the *Wigner–Seitz cell*, or *proximity cell*, which is constructed as follows. Choose a lattice point and draw lines joining all neighbouring points. Construct perpendicular planes bisecting these planes. The smallest volume thus enclosed is the Wigner–Seitz cell.

A total of 14 distinct types of three-dimensional lattices can be realised. These are known as the 14 *Bravais lattices*. For convenience, these are grouped into seven *lattice systems* according to their unit cell lengths and interaxial angles. The seven systems, with the number of their lattices given in parentheses, are triclinic (1), monoclinic (2), orthorhombic (4), tetragonal (2), cubic (3), trigonal (1) and hexagonal (1).

Here we will only consider one of the seven systems, namely, the *cubic system*, which is characterised by $a_1 = a_2 = a_3$ and $\hat{\mathbf{a}}_1 \cdot \hat{\mathbf{a}}_2 = \hat{\mathbf{a}}_2 \cdot \hat{\mathbf{a}}_3 = \hat{\mathbf{a}}_3 \cdot \hat{\mathbf{a}}_1 = 0$. The three different lattice types of this system are simple cubic (sc), face-centred cubic (fcc) and body-centred cubic (bcc). These are shown in [figure 1.1](#), where the cube is known as the *conventional cell* and a is called the cubic lattice constant.

SC

The primitive unit cell for the sc lattice is its conventional cell. The primitive translational vectors are

$$\begin{aligned} \mathbf{a}_1 &= a(1, 0, 0) \\ \mathbf{a}_2 &= a(0, 1, 0) \\ \mathbf{a}_3 &= a(0, 0, 1), \end{aligned} \quad (1.2)$$

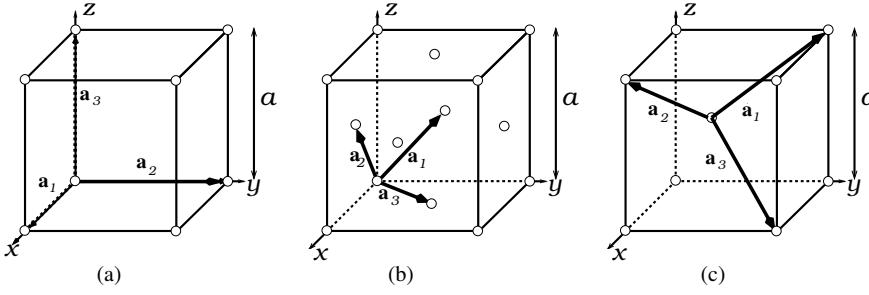


Figure 1.1 The cubic lattices: (a) SC, (b) FCC and (c) BCC. The cell shown is the conventional cell and a represents the cubic lattice constant. The primitive translation vectors $\mathbf{a}_1, \mathbf{a}_2$ and $\mathbf{a}_3$ are also shown.

where

$$(m_1, m_2, m_3) = m_1 \hat{\mathbf{x}} + m_2 \hat{\mathbf{y}} + m_3 \hat{\mathbf{z}}, \quad (1.3)$$

and $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ are unit vectors along the x, y and z directions in the cartesian system. The volume of the primitive cell, and hence volume per lattice point, is $\Omega = |\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)| = a^3$.

FCC

The primitive unit cell for the FCC lattice is defined by the three primitive translation vectors

$$\begin{aligned} \mathbf{a}_1 &= \frac{a}{2}(0, 1, 1) \\ \mathbf{a}_2 &= \frac{a}{2}(1, 0, 1) \\ \mathbf{a}_3 &= \frac{a}{2}(1, 1, 0). \end{aligned} \quad (1.4)$$

The volume of the unit cell is $\Omega = |\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)| = a^3/4$.

BCC

The primitive unit cell for the BCC lattice is defined by the three primitive translation vectors

$$\begin{aligned} \mathbf{a}_1 &= \frac{a}{2}(-1, 1, 1) \\ \mathbf{a}_2 &= \frac{a}{2}(1, -1, 1) \\ \mathbf{a}_3 &= \frac{a}{2}(1, 1, -1). \end{aligned} \quad (1.5)$$

The volume of the cell is $\Omega = |\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)| = a^3/2$.

1.2 RECIPROCAL LATTICE

The mathematical construction of a three-dimensional direct space lattice can be translated into an equivalent three-dimensional reciprocal space lattice. Such a mapping is achieved by using the relationship

$$\exp(i\mathbf{G} \cdot \mathbf{T}) = 1. \quad (1.6)$$

Corresponding to a direct translation vector $\mathbf{T}$, this relationship defines a *reciprocal translation vector* $\mathbf{G}$. The points generated by the vectors $\mathbf{G}$ form the *reciprocal lattice*. As direct lattice vectors $\mathbf{T}$ have the dimensions of [length], the reciprocal lattice vectors $\mathbf{G}$ have the dimensions of [1/length].

The reciprocal lattice is an essential concept for the study of crystalline solids and their diffraction properties. The space generated by reciprocal lattice points is called the *reciprocal space* (various other names are also commonly used, e.g. *G*-space, *q*-space, Fourier space, or momentum space).

The reciprocal lattice can be constructed, in conformity with the relationship in equation (1.6), both geometrically and mathematically. Geometrically, the reciprocal lattice is constructed as follows. Consider the lattice as a network of interpenetrating planes. Choose a lattice point as the origin and consider drawing normals to all possible planes in the direct lattice. If d is the distance of a plane from the origin, then represent this plane by a point at a distance $1/d$ on its normal. A collection of all such points produces the reciprocal lattice. Mathematically, the reciprocal lattice is defined by a set of vectors $\mathbf{G}$:

$$\mathbf{G} = m_1 \mathbf{b}_1 + m_2 \mathbf{b}_2 + m_3 \mathbf{b}_3, \quad (1.7)$$

where m_1, m_2 and m_3 are any integers and $\mathbf{b}_1, \mathbf{b}_2$ and $\mathbf{b}_3$ are the primitive translation vectors of the reciprocal lattice which are defined in terms of the primitive translation vectors $\mathbf{a}_1, \mathbf{a}_2$ and $\mathbf{a}_3$ of the direct lattice as follows:

$$\begin{aligned} \mathbf{b}_1 &= \frac{2\pi}{\Omega} (\mathbf{a}_2 \times \mathbf{a}_3) \\ \mathbf{b}_2 &= \frac{2\pi}{\Omega} (\mathbf{a}_3 \times \mathbf{a}_1) \\ \mathbf{b}_3 &= \frac{2\pi}{\Omega} (\mathbf{a}_1 \times \mathbf{a}_2), \end{aligned} \quad (1.8)$$

where $\Omega = |\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)|$ is the volume of the primitive unit cell of the direct lattice. It can be easily shown that $\mathbf{G} \cdot \mathbf{T} = 2\pi N$, where N is an integer, so that equation (1.6) is verified. The volume of the parallelepiped defined by $\mathbf{b}_1, \mathbf{b}_2$ and $\mathbf{b}_3$ is $\Omega_{rl} = |\mathbf{b}_1 \cdot (\mathbf{b}_2 \times \mathbf{b}_3)| = (2\pi)^3 / \Omega$. It can be easily verified that a reciprocal lattice vector $\mathbf{G}_{hkl} = h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3$ is perpendicular to a plane in the direct lattice with Miller indices (hkl) .

SC

Using equations (1.2) and (1.8), the primitive translation vectors of the reciprocal lattice of the sc lattice are

$$\begin{aligned} \mathbf{b}_1 &= \frac{2\pi}{a} (1, 0, 0) \\ \mathbf{b}_2 &= \frac{2\pi}{a} (0, 1, 0) \\ \mathbf{b}_3 &= \frac{2\pi}{a} (0, 0, 1). \end{aligned} \quad (1.9)$$

The volume of the parallelepiped defined by $\mathbf{b}_1, \mathbf{b}_2$ and $\mathbf{b}_3$ is $\Omega_{rl} = (2\pi/a)^3$.

FCC

The primitive translation vectors of the reciprocal lattice of the FCC direct space lattice are, using equations (1.4) and (1.8),

$$\begin{aligned} \mathbf{b}_1 &= \frac{2\pi}{a} (-1, 1, 1) \\ \mathbf{b}_2 &= \frac{2\pi}{a} (1, -1, 1) \\ \mathbf{b}_3 &= \frac{2\pi}{a} (1, 1, -1). \end{aligned} \quad (1.10)$$

The volume of the parallelepiped defined by $\mathbf{b}_1, \mathbf{b}_2$ and $\mathbf{b}_3$ is $\Omega_{rl} = 4(2\pi/a)^3$.

BCC

The primitive translation vectors of the reciprocal lattice of the BCC direct space lattice are

$$\begin{aligned} \mathbf{b}_1 &= \frac{2\pi}{a}(0, 1, 1) \\ \mathbf{b}_2 &= \frac{2\pi}{a}(1, 0, 1) \\ \mathbf{b}_3 &= \frac{2\pi}{a}(1, 1, 0). \end{aligned} \quad (1.11)$$

The volume of the primitive cell of the reciprocal lattice is $\Omega_{rl} = 2(2\pi/a)^3$.

Notice that the reciprocal lattice of the FCC direct space lattice is BCC, and vice versa. The reciprocal lattice of the sc direct space lattice remains sc. It can be verified that the reciprocal lattice of the reciprocal lattice is the original direct space lattice.

1.3 THE BRILLOUIN ZONE

It is customary in solid-state physics to choose the primitive unit cell of the reciprocal lattice as its central Wigner–Seitz cell, and not the parallelepiped defined by the primitive translation vectors $\mathbf{b}_1, \mathbf{b}_2$ and $\mathbf{b}_3$. The central Wigner–Seitz cell of a reciprocal lattice is known as its *first Brillouin zone* (or simply the Brillouin zone). The Wigner–Seitz cells of a reciprocal lattice outer to the central Wigner–Seitz cell generate the second Brillouin zone, third Brillouin zone, etc. For example, the second Brillouin zone is the volume enclosed between the central and the next outer Wigner–Seitz cells, and so on. Each Brillouin zone has the volume equal to $\Omega_{BZ} = \Omega_{rl} = |\mathbf{b}_1 \cdot (\mathbf{b}_2 \times \mathbf{b}_3)|$.

One of the most significant features of the Brillouin zone concept is that, for specular reflection from a set of parallel planes perpendicular to a reciprocal lattice vector $\mathbf{G}$, an elastic scattering $\mathbf{q}' = \mathbf{q} + \Delta\mathbf{q}$ ($|\mathbf{q}'| = |\mathbf{q}|$) can be expressed as

$$\begin{aligned} \Delta\mathbf{q} &= \mathbf{G} \\ \text{or} \quad q^2 &= (\mathbf{q} - \mathbf{G})^2 \\ \text{or} \quad G^2 - 2\mathbf{q} \cdot \mathbf{G} &= 0 \\ \text{or} \quad \mathbf{q} \cdot \hat{\mathbf{G}} &= \frac{1}{2}|\mathbf{G}|. \end{aligned} \quad (1.12)$$

In other words, the projection of $\mathbf{q}$ along a shortest reciprocal lattice vector $\mathbf{G}$ lies on the Brillouin zone face. This relation provides an alternative definition of the Bragg reflection, or diffraction, condition $2d_{hkl} \sin \theta = n\lambda$ for a wave of wavelength $\lambda = 2\pi/q$ propagating at an angle θ with respect to a set of parallel planes whose interplanar spacing is given by $d_{hkl} = 2\pi/G_{hkl}$, and where n is an integer.

SC

A general reciprocal lattice vector of the sc lattice is

$$\begin{aligned} \mathbf{G} &= m_1\mathbf{b}_1 + m_2\mathbf{b}_2 + m_3\mathbf{b}_3 \\ &= \frac{2\pi}{a}(m_1, m_2, m_3), \end{aligned} \quad (1.13)$$

where m_i are any integers.

The shortest $\mathbf{G}$ s are $\pm(2\pi/a)\hat{x}, \pm(2\pi/a)\hat{y}, \pm(2\pi/a)\hat{z}$. Therefore, there are six perpendicular bisector planes at distances, from the origin, $\pm(\pi/a)\hat{x}, \pm(\pi/a)\hat{y}, \pm(\pi/a)\hat{z}$. These planes enclose a cube of volume $(2\pi/a)^3$, which is the first Brillouin zone shown in [figure 1.2\(a\)](#).

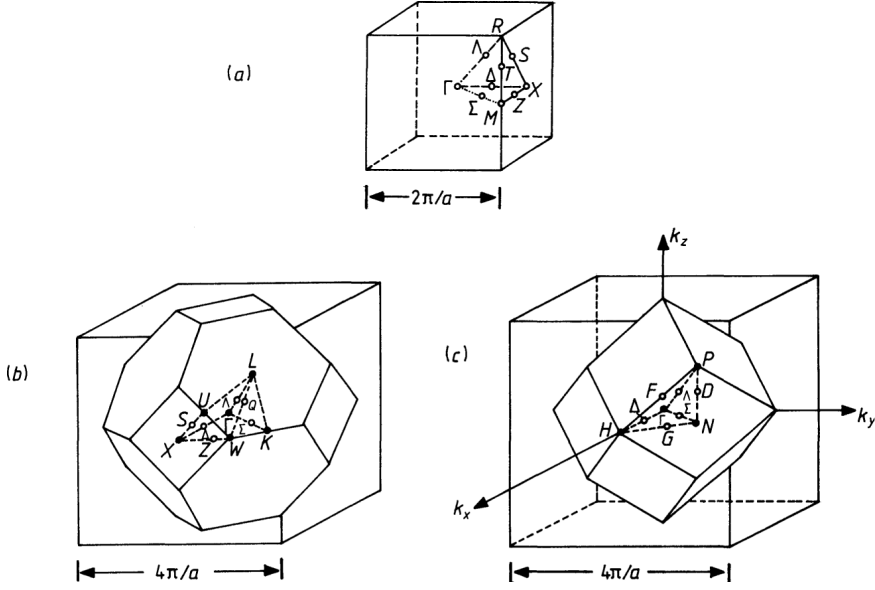


Figure 1.2 The Brillouin zone for cubic lattices: (a) SC, (b) FCC and (c) BCC. The irreducible part of the zone is also shown. (The shape of the Brillouin zone of the BCC lattice is identical to the shape of the Jones zone for the diamond/zincblende structure crystals (see [section 1.8](#)).

BCC

A general reciprocal lattice vector of the FCC lattice is

$$\begin{aligned}\mathbf{G} &= m_1\mathbf{b}_1 + m_2\mathbf{b}_2 + m_3\mathbf{b}_3 \\ &= \frac{2\pi}{a}(-m_1 + m_2 + m_3, m_1 - m_2 + m_3, m_1 + m_2 - m_3),\end{aligned}\quad (1.14)$$

where m_i are any integers.

The shortest $\mathbf{G}$ s are the eight vectors $(2\pi/a)(\pm 1, \pm 1, \pm 1)$. The next shortest $\mathbf{G}$ s are the six vectors $\pm(2\pi/a)(2, 0, 0), \pm(2\pi/a)(0, 2, 0), \pm(2\pi/a)(0, 0, 2)$. The Brillouin zone enclosed by the bisector planes perpendicular to these 14 vectors is a *truncated octahedron*, shown in [figure 1.2\(b\)](#).

BCC

A general reciprocal lattice vector of the BCC lattice is

$$\mathbf{G} = \frac{2\pi}{a}(m_2 + m_3, m_3 + m_1, m_1 + m_2),\quad (1.15)$$

where m_i are any integers.

The shortest $\mathbf{G}$ s are the 12 vectors $(2\pi/a)(0, \pm 1, \pm 1), (2\pi/a)(\pm 1, 0, \pm 1), (2\pi/a)(\pm 1, \pm 1, 0)$. The Brillouin zone defined by the bisector planes perpendicular to these planes is a regular 12-faced figure, a *rhombic dodecahedron*, as shown in [figure 1.2\(c\)](#).

1.4 CRYSTAL STRUCTURE

A crystal, or crystalline solid, is a regular three-dimensional arrangement of *atoms*. The concept of a lattice is useful in describing the arrangement of atoms in a crystal. For this purpose, we identify a

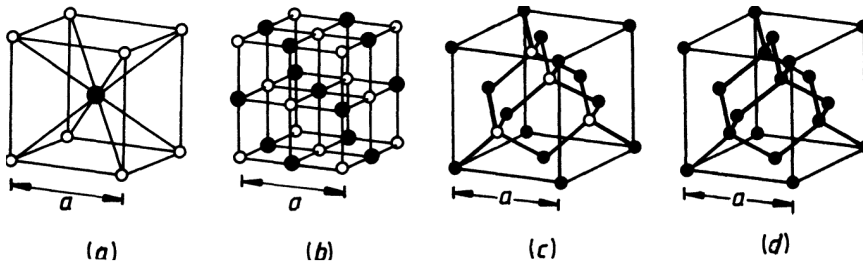


Figure 1.3 Some examples of crystal structure with cubic lattices: (a) CsCl structure with SC lattice, (b) NaCl structure with FCC lattice, (c) zincblende structure with FCC lattice, (d) diamond structure with FCC lattice. a denotes the cubic lattice constant.

to every lattice point. Thus a crystal has the repeat motif of its underlying lattice. A *crystal structure* is defined by the details of its lattice and basis:

$$\text{crystal structure} = \text{lattice} + \text{basis}$$

i.e. by the details of the parallelepiped of the primitive unit cell of its lattice and the distribution of atoms inside the unit cell.

Examples of crystal structures

We will consider a few examples of crystal structures based on the cubic lattice system.

Crystal structure with the sc lattice

The most common crystal structure with the sc lattice is the CsCl structure. In this structure, the basis consists of two atoms (one molecule of CsCl) with Cs at $(0,0,0)$ and Cl at $a(1/2, 1/2, 1/2)$ (or vice versa), where a is the cubic lattice constant. This structure is shown in [figure 1.3\(a\)](#). The number of nearest neighbours of each atom (the coordination number) of this structure is 8.

Crystal structures with the FCC lattice

We consider four structures with the FCC lattice.

The NaCl (or rocksalt) structure In this structure, which is also known as the alkali halide or rocksalt structure, each point of the FCC lattice is assigned a basis of one unit of NaCl: Na (group I atom) at $(0,0,0)$ and Cl (group VII atom) at $a(1/2, 1/2, 1/2)$ (or vice versa), as shown in [figure 1.3\(b\)](#). The coordination number of this structure is 6.

The CaF₂ structure

In this structure, the basis consists of Ca at $(0,0,0)$ and 2F at $\pm a(1/4, 1/4, 1/4)$.

The zincblende (sphalerite, or cubic ZnS) structure This is another type of crystal structure with the FCC lattice. As shown in [figure 1.3\(c\)](#), the basis consists of one atom at $(0,0,0)$ and another at $a(1/4, 1/4, 1/4)$. Semiconductors composed of atoms from group III and group V of the periodic table (known as III–V semiconductors) and many II–VI semiconductors crystallise in the zincblende structure. Each atom in this structure is tetrahedrally bonded to four atoms of the other type.

The diamond (or adamantine) structure The zincblende structure turns into the diamond crystal structure when the two basis atoms are identical. This crystal structure, therefore, is characterised by a centre-of-inversion at the midpoint between the two basis atoms. Crystals of group IV elements such as C (diamond), Si, Ge and α -Sn (grey tin) exhibit the diamond structure. In this structure, each atom is tetrahedrally bonded to four neighbours (of the same type of course), as shown in [figure 1.3\(d\)](#).

The diamond and zincblende structures are relatively open as the coordination number is only 4.

Many IV–IV, III–V and II–VI compounds can also be crystallised with the wurtzite structure (based on the hexagonal lattice), maintaining tetrahedral bonding. Some examples are SiC, GaN, ZnS, CdS, CdSe and ZnO.

Examples of crystal structures with the BCC lattice

An example of a crystal structure with the BCC lattice is provided by alkali halide metals such as Na and K. In this case, the basis consists of just one alkali atom, so that the structure of both the lattice and the crystal is the same: BCC. Another example is the SiF₄ crystal which also has the BCC lattice.

1.5 POINT GROUPS

Apart from its lattice translational symmetry, a crystal also shows two other types of symmetries: *point group* and *space group* symmetries.

An operation carried out about an axis passing through a point in space which brings the crystal structure into itself while leaving the point fixed is called a *point symmetry operation*. The collection of all point symmetry operations of a crystal forms a group which is called the *point symmetry group* (or simply *point group*). An allowed point symmetry operation can be one of the following types:

- (i) rotation about an n -fold axis: rotations through $(2\pi/n)$, with only $n=2,3,4$, or 6 are possible;
- (ii) reflection: reflection in a plane, called a mirror plane;
- (iii) inversion: an inversion through a point such that every point $\mathbf{r}$ is brought into $-\mathbf{r}$;
- (iv) rotation–reflection: rotation through $(2\pi/n)$ followed by a reflection in a plane perpendicular to the rotation axis;
- (v) rotation–inversion: rotation through $(2\pi/n)$ followed by an inversion about a point lying on the rotation axis.

Operations of type (i) are called proper rotations, while operations of types (ii)–(v) are called improper rotations. Thus a point group is a group of allowed proper and improper rotations of a lattice or crystal.

The total number of distinct point groups, corresponding to the seven lattice systems, is 32. These are generally known as the 32 *crystallographic point groups*. The total number of point groups for the cubic system is 5. Of particular interest to our discussion in this book are the diamond and zincblende cubic structures. The point group of the diamond structure is the full octahedral group O_h (or $m\bar{3}m$) with 48 operations. The point group of the zincblende structure is the tetrahedral group T_d (or $\bar{4}3m$) with 24 operations; as discussed earlier, the zincblende structure lacks the inversion symmetry present in the diamond structure. (Here we have used two commonly used notations: e.g. O_h is the *Schoenflies notation* (Schoenflies 1923) and $m\bar{3}m$ is the *International* (or *Hermann–Mauguin*) notation (Hermann 1930, Mauguin 1931).)

1.6 SPACE GROUPS

An operation about an axis passing through a point in space which leaves a crystal invariant, but which may involve moving the point within the primitive unit cell, is called a *space symmetry operation*. Such an operation can be represented by $\hat{O} = \{\mathcal{R} | \mathbf{t}\}$, where $\mathcal{R}$ represents a point symmetry operation and $\mathbf{t}$ is a *fractional translation* associated with $\mathcal{R}$,

$$\mathbf{t} = \mathbf{t}(\mathcal{R}) = u_1\mathbf{a}_1 + u_2\mathbf{a}_2 + u_3\mathbf{a}_3, \quad (1.16)$$

with $|u_i| < 1$. The collection of all space symmetry operations of a crystal forms a group which is called the *space symmetry group* (or simply *space group*). The group of operations $\{\mathcal{R} | 0\}$, called the *point group of the space group*, corresponds to one of the 32 crystallographic point groups.

Symmetry operations involving fractional translations are as follows.

(i) *Screw axes*: a screw axis operation is identified as a rotation followed by a fractional translation parallel to the rotation axis.

(ii) *Glide planes*: a glide plane operation is a fractional translation followed by a reflection in a plane containing the translation vector.

A space group operator $\{\mathcal{R} | \mathbf{t}\}$ corresponds to a coordinate transformation of the form

$$\mathbf{r}' = \mathcal{R}\mathbf{r} + \mathbf{t}, \quad (1.17)$$

where $\mathcal{R}$ can be represented by a 3×3 real orthogonal matrix in cartesian coordinates.

There are two types of space groups: *symmorphic* (or simple) and *non-symmorphic*. A symmorphic space group involves symmetry operations which do not involve any fractional translations. A non-symmorphic space group is specified by at least one operation which involves a fractional translation. There are a total of 230 (non magnetic) space groups, out of which 73 are symmorphic.

The space group of the diamond structure is the non-symmorphic group O_h^7 (or $Fd\bar{3}m$), involving 48 operations. The space group of the zincblende structure is the symmorphic group T_d^2 (or $F\bar{4}3m$), involving 24 operations. The operations of these space groups are listed in Table 1.1.

It should be mentioned that the fractional translations of a space group are dependent on the choice of origin. This is shown in Table 1.1 for two different choices of origin in the diamond and zincblende structure crystals. In particular, it can be noticed that the space group of the zincblende structure changes from symmorphic to non-symmorphic (i.e. non-zero fractional translations are introduced) when the origin is changed from one of the basis atoms to the midpoint between the two basis atoms.

1.7 SYMMETRY OF THE BRILLOUIN ZONE

If $\mathcal{R}$ is an operation in the point group of a lattice, then $\mathcal{R}^{-1}$ is also an operation of the same point group. Also, $\mathcal{R}^{-1}\mathbf{T}$ is a lattice translation if $\mathbf{T}$ is, so that

$$\mathcal{R}^{-1}\mathbf{T} \cdot \mathbf{G} = 2\pi N, \quad (1.18)$$

where N is some integer. As $\mathcal{R}$ represents an orthogonal transformation, i.e. $\mathcal{R}^T = \mathcal{R}^{-1}$, we can express

$$\begin{aligned} \mathbf{T} \cdot \mathcal{R}\mathbf{G} &= \sum_{ij} T_i \mathcal{R}_{ij} G_j \\ &= \sum_{ij} T_i \mathcal{R}_{ji}^{-1} G_j \\ &= \mathcal{R}^{-1}\mathbf{T} \cdot \mathbf{G} = 2\pi N, \end{aligned} \quad (1.19)$$

so that $\mathcal{R}\mathbf{G}$ must be a reciprocal lattice vector if $\mathbf{G}$ is. Therefore, it can be concluded that a reciprocal lattice has the same point group as its direct lattice. As a consequence of this result, it follows that the reciprocal lattice belongs to the same *system* as the direct lattice, although not necessarily the same *type*.

Although a primitive unit cell in the form of the parallelepiped defined by the vectors $\mathbf{a}_1, \mathbf{a}_2$ and $\mathbf{a}_3$ does not automatically display the point symmetry of its lattice, the symmetrical unit cell in the form of the Wigner–Seitz cell does. Therefore, the Brillouin zone of a lattice displays the point symmetry of the reciprocal lattice and hence the direct lattice. This means that, apart from the translational symmetry $f(\mathbf{q} + \mathbf{G}) = f(\mathbf{q})$, a function or property in the reciprocal space will also obey the point group symmetry of the direct lattice when $\mathbf{q}$ lies inside the Brillouin zone. Thus if the number of point group operations is M , then unique values of $f(\mathbf{q})$ exist only for $\mathbf{q}$ -values lying in a $(1/M)$ th part of the Brillouin zone. Such a part of the Brillouin zone is called its *irreducible part*.

Table 1.1

Space group operations of the diamond structure (O_h^7 group). The first 24 operations correspond to the zincblende structure (T_d^2 group).

j	$\{\mathcal{R} \mathbf{t}\}^{-1}\mathbf{r}$ origin at atomic site			$\{\mathcal{R} \mathbf{t}\}^{-1}\mathbf{r}$ origin midway between the two atoms		
1	x	y	z	x	y	z
2	$\bar{x}$	$\bar{y}$	z	$\bar{x} - \frac{a}{4}$	$\bar{y} - \frac{a}{4}$	z
3	x	$\bar{y}$	$\bar{z}$	x	$\bar{y} - \frac{a}{4}$	$\bar{z} - \frac{a}{4}$
4	$\bar{x}$	y	$\bar{z}$	$\bar{x} - \frac{a}{4}$	y	$\bar{z} - \frac{a}{4}$
5	y	$\bar{x}$	$\bar{z}$	y	$\bar{x} - \frac{a}{4}$	$\bar{z} - \frac{a}{4}$
6	$\bar{y}$	x	$\bar{z}$	$\bar{y} - \frac{a}{4}$	x	$\bar{z} - \frac{a}{4}$
7	$\bar{x}$	z	$\bar{y}$	$\bar{x} - \frac{a}{4}$	z	$\bar{y} - \frac{a}{4}$
8	$\bar{x}$	$\bar{z}$	y	$\bar{x} - \frac{a}{4}$	$\bar{z} - \frac{a}{4}$	y
9	$\bar{z}$	$\bar{y}$	x	$\bar{z} - \frac{a}{4}$	$\bar{y} - \frac{a}{4}$	x
10	z	$\bar{y}$	$\bar{x}$	z	$\bar{y} - \frac{a}{4}$	$\bar{x} - \frac{a}{4}$
11	$\bar{y}$	$\bar{x}$	z	$\bar{y} - \frac{a}{4}$	$\bar{x} - \frac{a}{4}$	z
12	$\bar{z}$	y	$\bar{x}$	$\bar{z} - \frac{a}{4}$	y	$\bar{x} - \frac{a}{4}$
13	x	$\bar{z}$	$\bar{y}$	x	$\bar{z} - \frac{a}{4}$	$\bar{y} - \frac{a}{4}$
14	y	x	z	y	x	z
15	z	y	x	z	y	x
16	x	z	y	x	z	y
17	z	x	y	z	x	y
18	y	z	x	y	z	x
19	z	$\bar{x}$	$\bar{y}$	z	$\bar{x} - \frac{a}{4}$	$\bar{y} - \frac{a}{4}$
20	$\bar{y}$	$\bar{z}$	x	$\bar{y} - \frac{a}{4}$	$\bar{z} - \frac{a}{4}$	x
21	$\bar{z}$	$\bar{x}$	y	$\bar{z} - \frac{a}{4}$	$\bar{x} - \frac{a}{4}$	y
22	$\bar{y}$	z	$\bar{x}$	$\bar{y} - \frac{a}{4}$	z	$\bar{x} - \frac{a}{4}$
23	$\bar{z}$	x	$\bar{y}$	$\bar{z} - \frac{a}{4}$	x	$\bar{y} - \frac{a}{4}$
24	y	$\bar{z}$	$\bar{x}$	y	$\bar{z} - \frac{a}{4}$	$\bar{x} - \frac{a}{4}$
25	$\bar{x} + \frac{a}{4}$	$\bar{y} + \frac{a}{4}$	$\bar{z} + \frac{a}{4}$	$\bar{x}$	$\bar{y}$	$\bar{z}$
26	$x + \frac{a}{4}$	$y + \frac{a}{4}$	$\bar{z} + \frac{a}{4}$	$x + \frac{a}{4}$	$y + \frac{a}{4}$	$\bar{z}$
27	$\bar{x} + \frac{a}{4}$	$y + \frac{a}{4}$	$z + \frac{a}{4}$	$\bar{x}$	$y + \frac{a}{4}$	$z + \frac{a}{4}$
28	$x + \frac{a}{4}$	$\bar{y} + \frac{a}{4}$	$z + \frac{a}{4}$	$x + \frac{a}{4}$	$\bar{y}$	$z + \frac{a}{4}$
29	$\bar{y} + \frac{a}{4}$	$x + \frac{a}{4}$	$z + \frac{a}{4}$	$\bar{y}$	$x + \frac{a}{4}$	$z + \frac{a}{4}$
30	$y + \frac{a}{4}$	$\bar{x} + \frac{a}{4}$	$z + \frac{a}{4}$	$y + \frac{a}{4}$	$\bar{x}$	$z + \frac{a}{4}$
31	$x + \frac{a}{4}$	$\bar{z} + \frac{a}{4}$	$y + \frac{a}{4}$	$x + \frac{a}{4}$	$\bar{z}$	$y + \frac{a}{4}$
32	$x + \frac{a}{4}$	$z + \frac{a}{4}$	$\bar{y} + \frac{a}{4}$	$x + \frac{a}{4}$	$z + \frac{a}{4}$	$\bar{y}$
33	$z + \frac{a}{4}$	$y + \frac{a}{4}$	$\bar{x} + \frac{a}{4}$	$z + \frac{a}{4}$	$y + \frac{a}{4}$	$\bar{x}$
34	$\bar{z} + \frac{a}{4}$	$y + \frac{a}{4}$	$x + \frac{a}{4}$	$\bar{z}$	$y + \frac{a}{4}$	$x + \frac{a}{4}$
35	$y + \frac{a}{4}$	$x + \frac{a}{4}$	$\bar{z} + \frac{a}{4}$	$y + \frac{a}{4}$	$x + \frac{a}{4}$	$\bar{z}$
36	$z + \frac{a}{4}$	$\bar{y} + \frac{a}{4}$	$x + \frac{a}{4}$	$z + \frac{a}{4}$	$\bar{y}$	$x + \frac{a}{4}$
37	$\bar{x} + \frac{a}{4}$	$z + \frac{a}{4}$	$y + \frac{a}{4}$	$\bar{x}$	$z + \frac{a}{4}$	$y + \frac{a}{4}$
38	$\bar{y} + \frac{a}{4}$	$\bar{x} + \frac{a}{4}$	$\bar{z} + \frac{a}{4}$	$\bar{y}$	$\bar{x}$	$\bar{z}$
39	$\bar{z} + \frac{a}{4}$	$\bar{y} + \frac{a}{4}$	$\bar{x} + \frac{a}{4}$	$\bar{z}$	$\bar{y}$	$\bar{x}$
40	$\bar{x} + \frac{a}{4}$	$\bar{z} + \frac{a}{4}$	$\bar{y} + \frac{a}{4}$	$\bar{x}$	$\bar{z}$	$\bar{y}$
41	$\bar{z} + \frac{a}{4}$	$\bar{x} + \frac{a}{4}$	$\bar{y} + \frac{a}{4}$	$\bar{z}$	$\bar{x}$	$\bar{y}$
42	$\bar{y} + \frac{a}{4}$	$\bar{z} + \frac{a}{4}$	$\bar{x} + \frac{a}{4}$	$\bar{y}$	$\bar{z}$	$\bar{x}$
43	$\bar{z} + \frac{a}{4}$	$x + \frac{a}{4}$	$y + \frac{a}{4}$	$\bar{z}$	$x + \frac{a}{4}$	$y + \frac{a}{4}$
44	$y + \frac{a}{4}$	$z + \frac{a}{4}$	$\bar{x} + \frac{a}{4}$	$y + \frac{a}{4}$	$z + \frac{a}{4}$	$\bar{x}$
45	$z + \frac{a}{4}$	$x + \frac{a}{4}$	$\bar{y} + \frac{a}{4}$	$z + \frac{a}{4}$	$x + \frac{a}{4}$	$\bar{y}$
46	$y + \frac{a}{4}$	$\bar{z} + \frac{a}{4}$	$x + \frac{a}{4}$	$y + \frac{a}{4}$	$\bar{z}$	$x + \frac{a}{4}$
47	$z + \frac{a}{4}$	$\bar{x} + \frac{a}{4}$	$y + \frac{a}{4}$	$z + \frac{a}{4}$	$\bar{x}$	$y + \frac{a}{4}$
48	$\bar{y} + \frac{a}{4}$	$z + \frac{a}{4}$	$x + \frac{a}{4}$	$\bar{y}$	$z + \frac{a}{4}$	$x + \frac{a}{4}$

Consider the diamond and zincblende structures. For both these structures, the Bravais lattice is FCC, and the point group is the octahedral group O_h with 48 operations. (Note that this is same as the point group of the diamond crystal.) Thus the irreducible part of the Brillouin zone of these crystal structures has the volume $\Omega_{BZ}/48$. One such irreducible segment is defined by the relations

$$\begin{aligned} 0 \leq q_x \leq q_y \leq q_z &\leq \frac{2\pi}{a} \\ q_x + q_y + q_z &\leq \frac{3}{2} \frac{2\pi}{a}. \end{aligned} \quad (1.20)$$

This is the *pentahedron* $\Gamma XULKW$ shown in [figure 1.2\(b\)](#) with five faces and six vertices at

$$\begin{aligned} \Gamma &= \frac{2\pi}{a}(0,0,0) & X &= \frac{2\pi}{a}(1,0,0) \\ U &= \frac{2\pi}{a}\left(1, \frac{1}{4}, \frac{1}{4}\right) & L &= \frac{2\pi}{a}\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right) \\ K &= \frac{2\pi}{a}\left(\frac{3}{4}, \frac{3}{4}, 0\right) & W &= \frac{2\pi}{a}\left(1, \frac{1}{2}, 0\right). \end{aligned} \quad (1.21)$$

The irreducible part of the Brillouin zone of the BCC lattice is the *tetrahedron* ΓHNP shown in [figure 1.2\(c\)](#) which has the volume $\Omega_{BZ}/48$, and where

$$\begin{aligned} \Gamma &= \frac{2\pi}{a}(0,0,0) & H &= \frac{2\pi}{a}(1,0,0) \\ N &= \frac{2\pi}{a}\left(\frac{1}{2}, \frac{1}{2}, 0\right) & P &= \frac{2\pi}{a}\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right). \end{aligned} \quad (1.22)$$

The irreducible part of the Brillouin zone for the sc lattice is the *tetrahedron* ΓXMR of volume $\Omega_{BZ}/48$, shown in [figure 1.2\(a\)](#), with its vertices at

$$\begin{aligned} \Gamma &= \frac{2\pi}{a}(0,0,0) & X &= \frac{2\pi}{a}\left(\frac{1}{2}, 0, 0\right) \\ M &= \frac{2\pi}{a}\left(\frac{1}{2}, \frac{1}{2}, 0\right) & R &= \frac{2\pi}{a}\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right). \end{aligned} \quad (1.23)$$

1.8 JONES ZONE

The first Brillouin zone, which is mathematically defined from the translational symmetry of the lattice, can accommodate in one energy band up to two valence electrons per primitive cell of the crystal. The usual concept of Brillouin zone boundaries is that they act as Bragg reflection planes for waves travelling inside the crystal. However, for a crystal with more than two valence electrons per primitive unit cell, the highest occupied electron energy surface (the Fermi surface) extends beyond the boundaries of the first Brillouin zone and, within the nearly-free-electron model, no splitting of the electron levels at the boundaries of the zone is expected. In other words, the boundaries of the first Brillouin zone do not act as Bragg reflection planes. In such a case, it is therefore useful to construct a larger zone, known as the *Jones zone*, with a volume just sufficient to accommodate all the valence electrons in the primitive unit cell. The faces of such a zone act as Bragg reflection planes, leading to discontinuities in electron energy levels.

For the diamond and zincblende structures, the Jones zone is constructed by the planes bisecting the 12 reciprocal lattice vectors of type (220): $(2\pi/a)(\pm 1, \pm 1, 0)$, $(2\pi/a)(0, \pm 1, \pm 1)$, $(2\pi/a)(\pm 1, 0, \pm 1)$. The shape of this zone is identical to the first Brillouin zone of the BCC lattice (shown in [figure 1.2\(c\)](#)): a rhombic dodecahedron.

We will use the concept of the Jones zone of the diamond structure in [section 3.3.2.3](#).

1.9 SURFACE BRILLOUIN ZONE

Surfaces of solids are the subject of a great deal of interest to physicists and chemists. The importance of studying semiconductor surfaces lies in their technological usage. To study electronic, optical or vibrational properties of a solid surface, it is essential to develop the concepts of surface periodicity and surface Brillouin zone.

Mathematically, we may construct the surface Brillouin zone by identifying the periodicity in the surface plane and introducing an artificial periodicity in the direction perpendicular to the surface. For example, for the case of the ideal zincblende (110) surface whose unit mesh is shown in [figure 1.4\(a\)](#), we choose

$$\begin{aligned} \mathbf{a}_1 &= \frac{a}{2}(-1, 1, 0) \\ \mathbf{a}_2 &= a(0, 0, 1) \\ \mathbf{a}_3 &= c(1, 1, 0), \end{aligned} \quad (1.24)$$

where $\mathbf{a}_1$ and $\mathbf{a}_2$ are the primitive translation vectors in the surface plane, and the vector perpendicular to the surface $\mathbf{a}_3$ is constructed with c arbitrarily related to the lattice constant a . In this particular case, the basis vectors define a parallelepiped of dimensions $(a/\sqrt{2}) \times a \times c\sqrt{2}$. As now

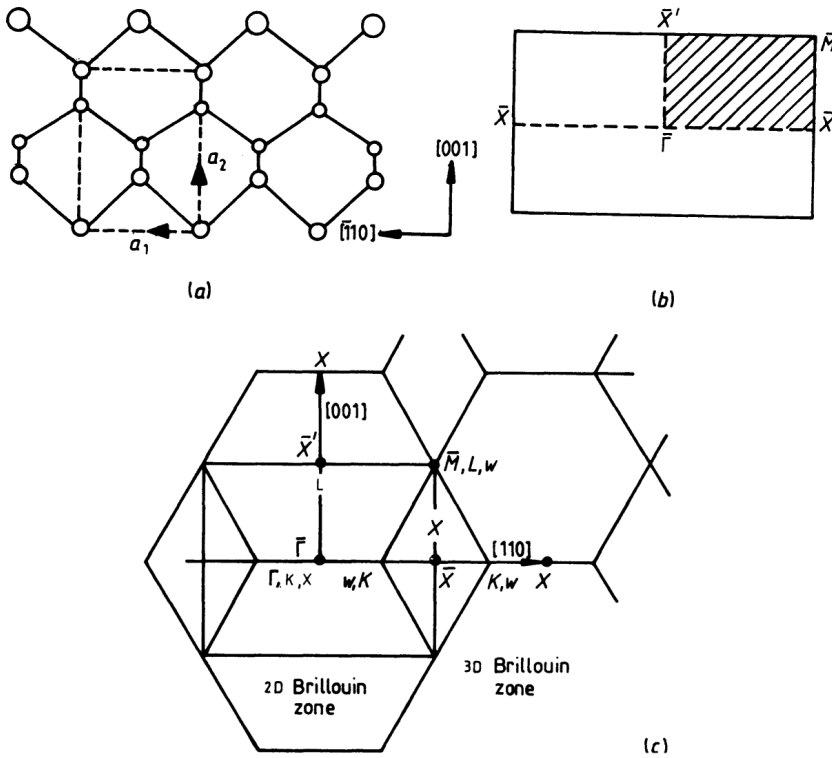


Figure 1.4 (a) The (110) unit mesh for the zincblende structure. Smaller dots represent atoms in the sub-surface layer. (b) The corresponding surface Brillouin zone. The symmetry points are labelled and the irreducible segment is shown by the hatched area. (c) Projection of the zincblende bulk Brillouin zone onto the (110) surface Brillouin zone. The back projection of bulk symmetry points is represented by small capital letters and points not projected are labelled by larger capital letters.

we are dealing with a periodic system, the procedure described in [section 1.2](#) can be used to construct the reciprocal lattice. The primitive translation vectors of the reciprocal lattice in the present example are

$$\begin{aligned} \mathbf{b}_1 &= \frac{2\pi}{\Omega}(\mathbf{a}_2 \times \mathbf{a}_3) = \frac{2\pi}{|\mathbf{a}_1|} \hat{\mathbf{a}}_1 \\ \mathbf{b}_2 &= \frac{2\pi}{\Omega}(\mathbf{a}_3 \times \mathbf{a}_1) = \frac{2\pi}{|\mathbf{a}_2|} \hat{\mathbf{a}}_2 \\ \mathbf{b}_3 &= \frac{2\pi}{\Omega}(\mathbf{a}_1 \times \mathbf{a}_2) = \frac{2\pi}{|\mathbf{a}_3|} \hat{\mathbf{a}}_3, \end{aligned} \quad (1.25)$$

where $\Omega = |\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)|$ is used. It should be mentioned that the vector $\mathbf{a}_3$ is only introduced to facilitate the construction of the reciprocal lattice using the concept described in [section 1.2](#). The surface reciprocal lattice is defined by setting the vector $\mathbf{b}_3$ to zero. Obviously the primitive unit cell of the reciprocal lattice of the surface is the area defined by $\mathbf{b}_1$ and $\mathbf{b}_2$.

A general vector of the surface reciprocal lattice is

$$\mathbf{G}_{\parallel} = m_1 \mathbf{b}_1 + m_2 \mathbf{b}_2 = \{m_1, m_2\}, \quad (1.26)$$

where m_i are any integers. For the present case, the first two sets of shortest $\mathbf{G}_{\parallel}$ s are simply $\pm \mathbf{b}_1$ and $\pm \mathbf{b}_2$. Therefore, the corresponding surface Brillouin zone is the area determined by the bisector planes perpendicular to these vectors. [figure 1.4\(b\)](#) shows the surface Brillouin zone for the (110) surface of the zincblende structure. This surface Brillouin zone exhibits the C_{2v} point group symmetry of the (110) surface. As this point group has four operations, the irreducible part of the surface Brillouin zone has a volume ($\frac{1}{4}$)th of the zone. This is the area $\bar{\Gamma}\bar{X}\bar{M}\bar{X}'$ shown in [figure 1.4\(b\)](#), where

$$\begin{aligned} \bar{\Gamma} &= \{0, 0\} & \bar{X}' &= \{0, \frac{1}{2}\} \\ \bar{X} &= \{\frac{1}{2}, 0\} & \bar{M} &= \{\frac{1}{2}, \frac{1}{2}\}, \end{aligned} \quad (1.27)$$

with $\{m_1, m_2\}$ defined in equation (1.26).

Geometrically, the surface Brillouin zone can be obtained by projecting the bulk Brillouin zone along the direction perpendicular to the surface. [Figure 1.4\(c\)](#) shows the projection of the zincblende bulk Brillouin zone onto the (110) surface Brillouin zone. The following projections of the symmetry points of the bulk Brillouin zone onto the surface Brillouin zone can be identified: $\Gamma, \kappa, x \rightarrow \bar{\Gamma}$; $x \rightarrow \bar{X}$; $L \rightarrow \bar{X}'$; $L, w \rightarrow \bar{M}$.

The surface Brillouin zones for two different periodicities in the (111) plane of the diamond structure, together with their irreducible segments, are shown in [figure 1.5](#).

1.10 MATRIX REPRESENTATIONS OF POINT GROUPS

The symmetry operations of a point group can be represented by square matrices (real or complex) which obey the multiplication table of the point group. The number of rows or columns in such a matrix is called the *dimension* of the representation. As can be realised, an infinite number of matrix representations of a point group is possible. However, for a given point group, there exists a set of fundamental representations, called *inequivalent irreducible representations*, from which *all* other representations can be obtained. Stated mathematically, if a representation for a point symmetry operation $\mathcal{R}$ could be expressed in the block form

$$D(\mathcal{R}) = \begin{pmatrix} D_1(\mathcal{R}) & 0 \\ 0 & D_2(\mathcal{R}) \end{pmatrix} \quad (1.28)$$

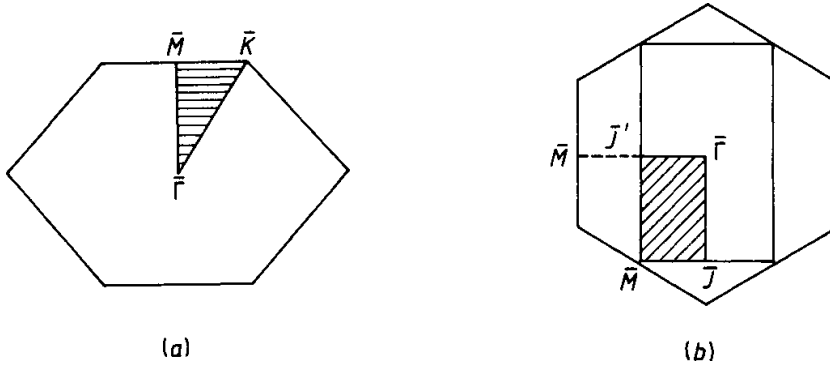


Figure 1.5 Brillouin zone for the (111) surface of the diamond/zincblende structure: (a) surface periodicity- 1×1 ; (b) surface periodicity- 2×1 .

then $D(\mathcal{R})$ is said to be *reducible*. If $D_i(\mathcal{R})$ are diagonal matrices so that their further reduction is not possible, then these are called *irreducible representations*.

An important quantity for an irreducible representation is its character which is the trace of its matrix

$$\chi_i(\mathcal{R}) = \sum_{\mu} D_{\mu\mu}^{(i)}(\mathcal{R}). \quad (1.29)$$

The characters of two different irreducible representations satisfy the orthogonality relation

$$\sum_{\mathcal{R}} \chi_i^*(\mathcal{R}) \chi_j(\mathcal{R}) = h \delta_{ij}, \quad (1.30)$$

where h is the number of elements in the group. Another important relation is

$$\sum_i l_i^2 = h, \quad (1.31)$$

where l_i is the dimensionality of the i th irreducible representation. Any two representations which have the same character are equivalent within a similarity transformation and are said to belong to the same *class*. Thus it follows that *the number of inequivalent irreducible representations of a group is equal to the number of classes in the group*.

There are three widely used notations for matrix representations of point groups: Bethe's notation $\Gamma_1, \Gamma_2, \dots, \Gamma_5$ (Bethe 1929); the BSW notation $\Gamma_1, \Gamma_2, \Gamma_{12}, \Gamma_{15}, \Gamma_{25}, \dots$ (Bouckaert *et al* 1936); and the EWK notation A, B, E, T (Eyring *et al* 1940). In this book, we will use all these notations. To make the reader familiar with these notations, we list here the irreducible representations of the point groups O_h , T_d , D_{2d} and C_{2v} . The point group $O_h(m\bar{3}m)$ has 10 irreducible representations. In the BSW (EWK) notation, there are 4 one-dimensional irreducible representations: $\Gamma_1(A_{1g})$, $\Gamma_{1'}(A_{1u})$, $\Gamma_2(A_{2g})$, $\Gamma_{2'}(A_{2u})$; 2 two-dimensional irreducible representations: $\Gamma_{12}(E_g)$, $\Gamma_{12'}(E_u)$; and 4 three-dimensional irreducible representations: $\Gamma_{15'}(T_{1g})$, $\Gamma_{15}(T_{1u})$, $\Gamma_{25'}(T_{2g})$, $\Gamma_{25}(T_{2u})$. The point group $T_d(\bar{4}3m)$ has five irreducible representations: in the (BSW, EWK, Bethe) notation, these are $(\Gamma_1, A_1, \Gamma_1)$, $(\Gamma_2, A_2, \Gamma_2)$, $(\Gamma_{12}, E, \Gamma_3)$, $(\Gamma_{25}, T_1, \Gamma_4)$ and $(\Gamma_{15}, T_2, \Gamma_4)$. In the Bethe (EWK) notation, the five irreducible representations of the point group $D_{2d}(\bar{4}2m)$ are $\Gamma_1(A_1)$, $\Gamma_2(A_2)$, $\Gamma_3(B_1)$, $\Gamma_4(B_2)$ and $\Gamma_5(E)$. The four irreducible representations of the point group C_{2v} in the Bethe (EWK) notation are $\Gamma_1(A_1)$, $\Gamma_3(A_2)$, $\Gamma_2(B_1)$ and $\Gamma_4(B_2)$. In the above list, A_i are one-dimensional representations with character $+1$, B_i are one-dimensional representations with character -1 , and the subscripts g and u represent the parity of the representation under inversion: $g = \text{gerade} = \text{even}$, $u = \text{ungerade} = \text{odd}$. The irreducible representations of other point groups can be found in many standard textbooks on group theory.

1.11 EFFECT OF SPACE GROUP OPERATIONS ON PLANE WAVES

It was discussed earlier that the space group of a crystal is its full symmetry group. A space group operator is expressed as $\{\mathcal{R} | \mathbf{t}\}$, where $\mathcal{R}$ is either a proper or an improper rotation operator and $\mathbf{t}$ represents a fractional translation associated with $\mathcal{R}$. (The group of lattice translation vectors, which has operations of the type $\{0|\mathbf{T}\}$, is also a part of space group but we have excluded this from our discussion.) Using equation (1.17), it can be verified that the effect of two successive operations is given by

$$\{\mathcal{R}' | \mathbf{t}'\}\{\mathcal{R} | \mathbf{t}\} = \{\mathcal{R}'\mathcal{R} | \mathcal{R}'\mathbf{t} + \mathbf{t}'\}. \quad (1.32)$$

The identity element of the space group is $\{E | 0\}$. Furthermore, the inverse of the operator $\{\mathcal{R} | \mathbf{t}\}$ is

$$\{\mathcal{R} | \mathbf{t}\}^{-1} = \{\mathcal{R}^{-1} | -\mathcal{R}^{-1}\mathbf{t}\}. \quad (1.33)$$

This can be verified by showing, with the help of equation (1.32), that

$$\{\mathcal{R} | \mathbf{t}\}\{\mathcal{R} | \mathbf{t}\}^{-1} = \{E | 0\}. \quad (1.34)$$

The effect of an operation $\hat{O} = \{\mathcal{R} | \mathbf{t}\}$ on a function or property of a crystal at a point in its unit cell is given as

$$\hat{O}f(\mathbf{r}) = f(\{\mathcal{R} | \mathbf{t}\}^{-1}\mathbf{r}) \quad (1.35)$$

$$= f(\mathcal{R}^{-1}\mathbf{r} - \mathcal{R}^{-1}\mathbf{t}). \quad (1.36)$$

Let us consider $f(\mathbf{r})$ as a plane wave $\exp(i\mathbf{q} \cdot \mathbf{r})$. Then

$$\begin{aligned} \hat{O}\exp(i\mathbf{q} \cdot \mathbf{r}) &= \exp(i\mathbf{q} \cdot \{\mathcal{R} | \mathbf{t}\}^{-1}\mathbf{r}) \\ &= \exp[i\mathbf{q} \cdot (\mathcal{R}^{-1}\mathbf{r} - \mathcal{R}^{-1}\mathbf{t})] \\ &= \exp(-i\mathbf{q} \cdot \mathcal{R}^{-1}\mathbf{t}) \exp(i\mathbf{q} \cdot \mathcal{R}^{-1}\mathbf{r}) \\ &= \exp(-i\mathcal{R}\mathbf{q} \cdot \mathbf{t}) \exp(i\mathcal{R}\mathbf{q} \cdot \mathbf{r}) \\ &= \gamma \exp(i\mathcal{R}\mathbf{q} \cdot \mathbf{r}), \end{aligned} \quad (1.37)$$

where we have used the orthogonal nature of the matrix representing $\mathcal{R}$. Thus upon the application of a space group operation, a plane wave characterised by a wave vector $\mathbf{q}$ is transformed, to within a phase factor $\gamma \equiv \exp(-i\mathcal{R}\mathbf{q} \cdot \mathbf{t})$, into another plane wave which is characterised by a wave vector $\mathbf{q}' = \mathcal{R}\mathbf{q}$. The phase factor γ is ± 1 for all operations of a symmorphic space group. In general, however, for the cubic system γ takes values ± 1 , or $\pm i$.

All operations $\mathcal{R}$ which leave $\mathbf{q}$ invariant or carry it into an equivalent point, i.e. when

$$\mathbf{q}' = \mathcal{R}\mathbf{q} = \mathbf{q} + \mathbf{G} \quad (1.38)$$

form a group $C(\mathbf{q})$, which is called the *group of the wave vector* $\mathbf{q}$. This group is a subgroup of the full space group of the crystal under consideration. The set of distinct $\mathbf{q}'$ (taking $\mathbf{q}$ to new positions) generated by the application of all the group operations is called the 'star' of $\mathbf{q}$. In general, the more symmetric the point $\mathbf{q}$, the fewer the number of vectors in its star. The product of the order of the group of $\mathbf{q}$ and the number of vectors in its star equals the order of the full point group.

The irreducible representations of the group $C(\mathbf{q})$ can be used to obtain all the irreducible representations of the full space group. The reader is referred to the excellent review article by Koster (1957) for details on this topic.

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